Co-Optimizing District Energy Networks and Flexible AI Workloads: A Rolling-Horizon MILP Approach

ENERGY 291 Final Report

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1 Introduction

Increased electrification, particularly from large loads such as data centers, electric vehicles, and electrified heating/cooling systems, has exacerbated load matching concerns across the United States [1]. For data centers in particular, the sudden surge of generative AI development has introduced unprecedented challenges for system operators regarding the reliable and secure operation of the electric grid. Although historically treated by utilities as static, inflexible loads with flat demand profiles, modern data center facilities possess a unique opportunity to act as dynamic, flexible demand-side agents. The asynchronous nature of deep learning AI training offers the inherent operational flexibility to shift delay-tolerant computational tasks to optimized periods of the day.

Thus, by treating AI compute capacity as a Virtual Power Plant (VPP), data center operators can dynamically modulate their real-time power consumption in response to grid stress, price volatility, and marginal carbon emissions signals. This shift changes the narrative of data center demand from an infrastructure liability into a grid-stabilizing asset.

For this project, we developed a multi-period Mixed-Integer Linear Programming (MILP) optimization framework executed within a rolling-horizon Model Predictive Control (MPC) window. This multi-objective model co-optimizes virtual workload queue buffering alongside physical thermal energy storage systems. The objective is to maximize raw computational throughput while minimizing operational expenditures and grid-level carbon emissions.

The Stanford University campus ecosystem provides a unique opportunity for this thesis. Stanford operates a fully electrified district energy network via its Central Energy Facility (CEF), which utilizes large-scale thermal energy storage (TES) tanks and Heat Recovery Chillers (HRC) to meet campus heating and cooling demands [2]. By integrating empirical thermodynamic data from the CEF, public cloud bursting price traces, and an open-source hyper-scale AI training workload trace (scaled to match Stanford's localized capacity constraints), we model a thermally coupled, computationally flexible digital twin to address the escalating AI power crisis.

At Stanford specifically, research-level AI training consumes a massive and growing share of institutional resources. The 2025 Stanford Artificial Intelligence Report [3] demonstrated that the model scale of frontier AI architectures continues to grow exponentially, with training compute parameters doubling every five months and power demands doubling annually. The Stanford Research Computing Facility (SRCF) demonstrates this growth trajectory; its newest campus facility expansion brings localized capacity availability up to 6 MW of compute [4]. Managing volatile research compute job submissions within this 6 MW infrastructure constraint forms the core challenge of this project.

This modeling architecture, which is focused on the co-optimization of low-carbon district thermal systems with flexible compute demand, can serve as a blueprint for future large-load grid interconnections. The trifecta of accelerating AI, aging power infrastructure, and multi-year utility interconnection queues highlights the importance of this network for safely interconnecting and considering hyper-scale compute loads. Optimizing localized virtual storage and flexible data center scheduling directly mitigates the immediate pressure for capital-intensive transmission grid build-out. By meeting emerging computational loads through coordinated software scheduling for existing infrastructure, institutions can capture large cost savings, reduce carbon dioxide emissions, and accelerate project timelines.

2 Data

We construct a digital twin simulation that couples the computational dispatch of a 6 MW AI supercomputing cluster with the physical thermal energy storage (TES) operations of Stanford's district heating and cooling network. The core reasoning behind this simulation is to evaluate whether a predictive, queue-based workload scheduler can successfully leverage thermal flexibility to avoid expensive public cloud outsourcing and minimize grid-level carbon emissions. To construct this environment, we integrate three distinct empirical datasets, each aligned across an 8,760-hour annual horizon, to represent the Stanford campus ecosystem:

1. Stanford Central Energy Facility (CEF) Data: We utilize historical operational data from the Stanford CEF to define the baseline thermodynamic demands of the campus. Joe Stagner, whose team oversees CEF operations and who helped design Stanford's Central Energy Plant Optimization Model (CEPOM), provided us with a full year of hourly data (May 2025–May 2026) covering CEF electrical load [kWh], heating demand [mmBtu], cooling demand [ton-hours], heating and cooling water supply and return temperatures [F], heating tank thermal energy storage [mmBtu] and storage capacity utilization [%] with a given maximum capacity [mmBtu], cooling tank thermal energy storage [ton-hours] and storage capacity utilization [%] with a given maximum capacity [ton-hours]. This dataset also provides the physical boundary conditions for the system, including the HRC maximum cooling limits (2,500 tons), conventional chiller Coefficient of Performance (COP = 5.5), max TES tank flow rates [gpm], turndown limits [% and tons], and maximum electrical draw [kW]. We additionally received baseline load profiles detailing historical hourly electricity, heating, and cooling demand exclusive of the data center. To link the IT electrical compute load to the facility's thermal cooling demand, we applied standard data center efficiency metrics, specifically a verified Power Usage Effectiveness (PUE) multiplier derived from industry benchmarks [5]. Joe's CEF team also provided full-year hourly California Independent System Operator (CAISO) Day-Ahead Market electricity prices [$/MWh] for

Stanford's procured electricity, which we used to drive cost minimization.

Accurately including pricing information for energy charges was a major bottleneck in our research. Although Stanford collects a majority of its electricity through PPAs with off-site solar farms [6], it still must pay a "standby" demand charge to PG&E for the solar capacity in the event that the solar generation does not match demand [7]. This demand charge comes in addition to a standard volumetric charge. Although a major component of Stanford's cost structure, we omitted this payment for simplicity in our model and a lack of data. A more robust iteration of this optimization model would include demand charges.

2. NREL Cambium Emission Forecasts: To accurately penalize carbon-intensive grid draw within the objective function, we integrate the National Renewable Energy Laboratory (NREL) 2024 Cambium forecasts [8]. Specifically, we utilize the Levelized Long-Run Marginal Emission Rate (LRMER) data (λ_t) for the California region. This provides an hourly metric of induced CO_2 emissions ($kgCO_2$/MWh) associated with incremental grid load. This demonstrates the duck-curve dynamics of the CAISO grid. By comparing our optimized dispatch against historical business-as-usual operations using both the CEF and Cambium datasets, we quantified the resulting changes in cost, grid flexibility, and carbon savings.

3. Alibaba GPU Cluster Trace: To serve as the primary flexible load and to model the spiky nature of high-performance computing demand, we synthesize an 8,760-hour compute arrival trace (W_t) utilizing the publicly available Alibaba GPU Cluster Trace Program dataset [9]. This program provides sub-hourly metadata that allowed us to distinguish between latency-sensitive inference and delay-tolerant training tasks. We aggregated raw job, task, and sensor tables (including *cpu_usage* and *gpu_wrk_util*) to generate a unified time-series vector of power demand. The raw dataset was isolated to a 36,500 MWh annual payload and mapped sequentially over the 8,760-hour simulation window. This dataset serves to dynamically stress-test the 500 MWh virtual queueing mechanism against highly variable industrial AI workloads. However, a key limitation of this approach is that we were unable to obtain real-time GPU and CPU utilization logs from Stanford's on-campus Research Computing Facility (SRCF). Access to Stanford-native workload data, rather than using Alibaba cluster trace as a proxy, would have more accurately reflected the specific job mix and submission patterns of Stanford's cluster.

A table listing all major data inputs and their sources is included below in Table 1:

Table 1: Summary of Major Data Inputs

Variables and Units	Dataset	Source
Cooling demand (D^{chw}) [ton-hours], Heating demand (D^{hw}) [mmBtu], CHW/HW TES state of charge [%], Supply/return temperatures [°F]; May 2025–May 2026	Stanford CEF Operational Data	Joe Stagner, Stanford CEF / CEPOM
HRC cooling limits H^{cool}_{max}, H^{cool}_{min} [tons], electrical draw coefficients ρ_{hrc}, $\rho_{chiller}$ [MW/ton], conventional chiller COP $= 5.5$, max TES tank flow rates [gpm]	HRC & Chiller Parameters	Stanford CEF / Joe Stagner
Grid electricity price (π_t) [$/MWh]; 8,760-hour annual profile	CAISO Day-Ahead Electricity Prices	Stanford CEF / PG&E
Levelized LRMER (λ_t) [kgCO_2/MWh]; California region; 8,760-hour annual profile	NLR Cambium 2024 Forecasts	National Laboratory of the Rockies (NLR)
Natural gas boiler emissions intensity (ϵ_{boiler}) [kgCO_2/mmBtu]	EPA Stationary Combustion Emission Factors	U.S. Environmental Protection Agency
CPU usage, GPU utilization [%], job start/end times [s]; scaled to 36,500 MWh annual payload	Alibaba GPU Cluster Trace (2018)	Alibaba Cluster Trace Program
Power Usage Effectiveness ($\phi = 1.15$) [unitless]	PUE Benchmark	National Laboratory of the Rockies (NLR)

3 Methods

3.1 Problem Formulation

We formulate our digital twin framework as a deterministic, rolling-horizon Mixed-Integer Linear Program (MILP). All constraints are linear equalities and inequalities, and all decision variables are continuous and non-negative except for the binary HRC commitment variable $U_t \in \{0, 1\}$, which introduces the integer component. The objective is to co-optimize the operation of a 6 MW AI supercomputing cluster and the Stanford Central Energy Facility (CEF) by minimizing a weighted scalarized sum of operational costs, lifecycle carbon emissions, and AI workload queuing delays.

Let $\mathcal{T}$ represent the set of hourly timesteps in our optimization window. We pose the constrained linear optimization problem as follows:

$$\min \quad w_{cost}\left(\frac{C_{total}}{\eta_{cost}}\right) + w_{carbon}\left(\frac{E_{total}}{\eta_{carbon}}\right) + w_{speed}\left(\frac{Q_{total}}{\eta_{speed}}\right) \tag{1}$$

Where the total cost, C_{total} ($), total CO_2 emissions, E_{total} (metric tons of CO_2), and total queue backlog, Q_{total} (MWh) are the decision variables, defined over $\mathcal{T}$ as:

$$C_{total} = \sum_{t \in \mathcal{T}} \Big(G_t \cdot \pi_t + B_t \cdot c_{boiler} + O_t \cdot c_{cloud} \Big) \tag{2}$$

$$E_{total} = \frac{1}{1000} \sum_{t \in \mathcal{T}} \Big(G_t \cdot \lambda_t + B_t \cdot \epsilon_{boiler} \Big) \tag{3}$$

$$Q_{total} = \sum_{t \in \mathcal{T}} Q_t \tag{4}$$

subject to:

$$0 \le A_t \le A_{max}, \quad \forall t \in \mathcal{T} \tag{5}$$

$$0 \le Q_t \le Q_{max}, \quad \forall t \in \mathcal{T} \tag{6}$$

$$Q_t = Q_{t-1} + W_t - A_t - O_t, \quad \forall t \in \mathcal{T} \tag{7}$$

$$H^{cool}_{min} \cdot U_t \le H^{cool}_t \le H^{cool}_{max} \cdot U_t, \quad \forall t \in \mathcal{T} \tag{8}$$

$$H^{heat}_t = H^{cool}_t \cdot \gamma_{hrc}, \quad \forall t \in \mathcal{T} \tag{9}$$

$$S^{chw}_t = S^{chw}_{t-1} + H^{cool}_t + C_t - D^{chw}_t - (A_t \cdot \phi \cdot \rho), \quad \forall t \in \mathcal{T} \tag{10}$$

$$S^{hw}_t = S^{hw}_{t-1} + H^{heat}_t + B_t - D^{hw}_t, \quad \forall t \in \mathcal{T} \tag{11}$$

$$G_t = A_t + (H^{cool}_t \cdot \rho_{hrc}) + (C_t \cdot \rho_{chiller}), \quad \forall t \in \mathcal{T} \tag{12}$$

In this formulation, A_t is the local AI compute executed (MW), Q_t is the queue backlog (MWh), O_t is the emergency cloud outsource volume (MWh), and W_t represents the raw AI workload arrivals (MWh). G_t represents the total grid draw (MW), while π_t and λ_t denote the hourly grid electricity price (\$/MWh) and Long-Run Marginal Emission Rate (LRMER) ($kgCO_2$/MWh), respectively. The boiler heat output is given by B_t (mmBtu), with associated fuel cost c_{boiler} (\$/mmBtu) and emissions intensity ϵ_{boiler} ($kgCO_2$/mmBtu) [10]. Cloud outsourcing is priced at c_{cloud} (\$/MWh).

The binary variable $U_t \in \{0, 1\}$ encodes the on/off commitment status of the Heat Recovery Chiller (HRC), which operates between turndown limits H^{cool}_{min} and H^{cool}_{max} (tons). H^{cool}_t (tons) and H^{heat}_t (mmBtu) denote the HRC cooling and heat recovery outputs, respectively, linked by the recovery ratio γ_{hrc} (mmBtu/ton). The conventional chiller output is C_t (tons). The electrical draw coefficients ρ_{hrc} (MW/ton) and $\rho_{chiller}$ (MW/ton) convert cooling output to grid electricity consumption for the HRC and conventional chiller, respectively.

The chilled water and hot water TES tank states of charge are given by S^{chw}_t (ton-hours) and S^{hw}_t (mmBtu), driven by campus cooling and heating demand profiles D^{chw}_t (ton-hours) and D^{hw}_t (mmBtu). The AI cooling load coupled into the chilled water tank is computed via the Power Usage Effectiveness multiplier ϕ (PUE = 1.15) and the unit conversion factor $\rho = 284.345$ MW/ton.

The unitless scalar weights $w_{cost} = 0.35$, $w_{carbon} = 0.30$, and $w_{speed} = 0.35$ reflect a the desire to treat economics and computational throughput as co-equal primary objectives, while carbon emissions reduction is weighted as a strongly important but secondary consideration. The normalization constants η_{cost}, η_{carbon}, and η_{speed} are unitless fixed scaling denominators that map each objective term onto a comparable numerical range, preventing any single objective from dominating the scalarized sum due to differences in units or magnitude.

Equations 10 and 11 enforce the thermodynamic balance of the chilled water (S^{chw}) and hot water (S^{hw}) thermal energy storage (TES) tanks. Equation 12 enforces the facility-level electrical power balance, linking computational load, Heat Recovery Chiller (HRC) electrical draw, and conventional chiller electrical draw. The maximum AI capacity is bounded at $A_{max} = 6$ MW, while the maximum queue backlog is bounded at $Q_{max} = 500$ MWh, allowing a maximum theoretical delay of 120 hours (5 days).

3.2 Model Predictive Control Architecture

Unlike historical time-series clustering approaches, which group representative days to solve a single static optimization, we implemented a Model Predictive Control (MPC) architecture. The model optimizes over a 48-hour rolling prediction horizon to account for impending fluctuations in daily utility pricing and grid carbon intensity. However, only the decisions for the first 24 hours are committed to the historical state vector, after which the horizon rolls forward by one day.

We implemented this architecture using Julia. The MILP was formulated using JuMP (a mathematical optimization modeling language embedded in Julia) and solved with the HiGHS solver.

4 Results

To evaluate the efficacy of the Model Predictive Control (MPC) co-optimization framework, the 8,760-hour digital twin simulation was executed against a baseline "uncoordinated" scenario. The uncoordinated baseline assumes that AI workloads must be processed instantly upon arrival with zero allowable queuing delay, forcing excess capacity immediately into an external cloud equivalent (AWS Spot pricing at \$1,200/MWh) [11].

The results demonstrate profound improvements across peak grid strain, operational flexibility, thermal utility co-location, and bottom-line financial and environmental performance.

4.1 Annual Cost and Carbon Savings

The implementation of the flexible compute queue and rolling MPC optimization yielded large cumulative savings. At the end of the 8,760-hour simulation, the uncoordinated baseline incurred \$15,177,824 in total operational costs and induced 27,286 metric tons (MT) of carbon emissions. In contrast, the MPC model lowered total costs to $7,964,237 and emissions to 24,360 MT.

This represents an aggregate annual savings of $7,964,237 (a reduction of 52.5%) and an avoidance of 2,926 MT of CO_2 (a reduction of 10.7%). These cumulative divergence trajectories are visually confirmed in Figures 5 and 6, which illustrate how the MPC algorithmic intervention continually compounded savings hour-by-hour over the course of the year.

4.2 Peak Shaving and Load Duration Smoothing

Figure 1 presents the annual Load Duration Curve (LDC) for total facility grid draw, sorting the 8,760 hours of the year from highest electrical demand to lowest. As observed on the far left of the curve, both the uncoordinated baseline and the MPC model experience a narrow spike of extreme peak loads. The structural difference between the two curves emerges in the mid-to-low demand regime. The MPC model holds a roughly flat draw near 9.5 MW from approximately hour 3,500 through hour 6,500 before falling sharply to its minimum, while the uncoordinated baseline tapers more gradually across the same range. The shaded region at the right tail (hours 6,500–8,500) represents hours where the baseline actually exceeds MPC grid draw. This occurs as a result of a lack of compute deferral in the uncoordinated case. Without foresight, the baseline system services residual thermal and compute loads reactively during the low-demand hours, whereas the MPC has already discharged those loads during earlier, higher-draw windows. The net effect is not classical peak shaving, but rather a rightward compression of the distribution where the MPC consolidates consumption into deliberate mid-range hours and reaches its baseload floor sooner.

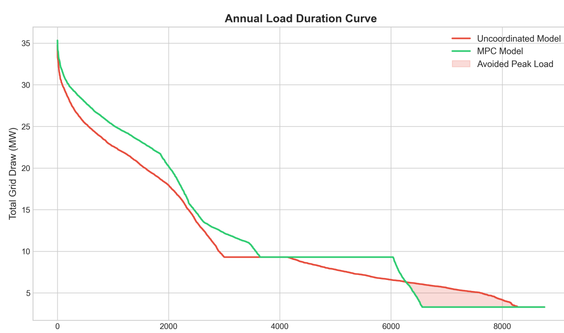

Figure 1: Load Duration Curve for Facility Draw

4.3 Multi-Objective Daily Shifting

The average 24-hour dispatch profile in Figure 2 illustrates the rolling horizon's ability to navigate CAISO grid pricing and NREL marginal emissions data (LRMER). Although the uncoordinated baseline is forced into high electrical draw symmetrically throughout the day based purely on random researcher submissions, the optimized MPC model actively suppresses grid draw during the late afternoon and early evening hours. By storing AI workloads in the queue during the expensive and carbon-intensive time period (hours 16–21), the solver shifts processing into the night and solar-dominant morning/mid-day periods, arbitraging both carbon and cost.

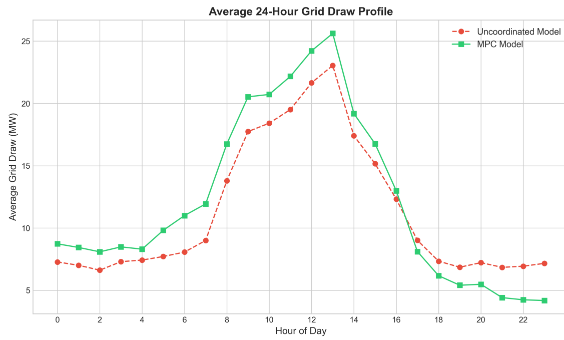

Figure 2: Average 24-Hour Dispatch Profile

4.4 Queuing Viability vs. Cloud Outsourcing

Figure 3 presents the annual time series of queue utilization under the MPC model alongside forced cloud outsourcing events. The MPC model's local queue backlog (orange curve) exhibits a distinctly episodic structure, cycling between near-zero values and sustained spikes reaching 400–480 MWh. The plateaus correspond to the model deliberately withholding compute job execution and banking work into the deep queue buffer. This stores deferred compute that can be discharged during grid-favorable windows. The uncoordinated baseline (red curve) generates small spikes of forced AWS cloud outsourcing throughout the year, remaining near zero in total energy volume but incurring continuous, expensive cloud procurement costs whenever local capacity is exceeded.

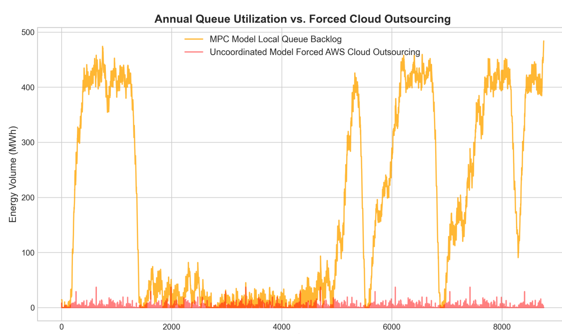

Figure 3: Queue Utilization vs. Forced Outsourcing

4.5 Operational Cost Abatement

The financial divergence between the uncoordinated baseline and the MPC model is substantial. Over the 8,760-hour simulation, the baseline model accumulated \$15,177,824 in total operational expenditures. A significant majority of this expense—over \$7.2 million—was driven by forced emergency cloud outsourcing, as the inflexible system continuously breached local capacity limits and defaulted to \$1,200/MWh spot pricing.

In contrast, the MPC model completely eliminated the need for external cloud bursting by utilizing the 500 MWh virtual queue as a computational shock absorber. Consequently, the optimized framework restricted annual costs strictly to local grid and heating operations. As depicted in Figure 4, this shift yields an absolute savings of $7,964,237.

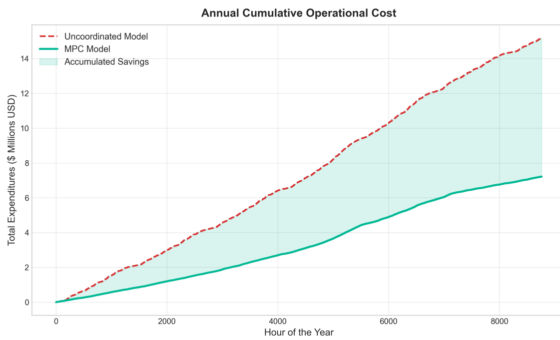

Figure 4: Cumulative Cost Advantage

4.6 Grid-Level Decarbonization

Beyond financial metrics, the MPC model actively optimizes for environmental sustainability by integrating NREL Cambium Long-Run Marginal Emission Rate (LRMER) forecasts into the objective function. The uncoordinated baseline, which is blind to dynamic grid carbon intensity, induced 27,286 Metric Tons (MT) of CO_2. This high footprint occurs because the baseline forces compute processing during peak evening hours, heavily relying on California's natural gas peaker plants.

By contrast, the MPC solver penalizes carbon-intensive grid periods, intentionally buffering compute workloads during these dirty hours and draining the queue when renewable penetration is highest. Figure 5 tracks this cumulative environmental benefit, revealing that the MPC model generated only 24,360 MT of CO_2. This temporal load shifting successfully averts 2,926 MT of carbon emissions.

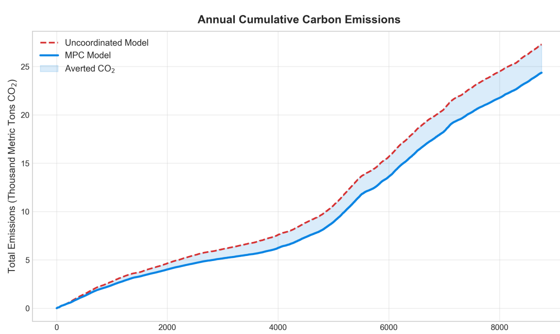

Figure 5: Emissions Footprint

5 Sensitivity Analysis

A critical element of the digital twin framework is evaluating its robustness against uncertain exogenous variables, specifically Service Level Agreement (SLA) constraints and cloud market volatility. To determine the boundary conditions of the primary simulation, a sensitivity analysis was conducted across researcher delay tolerances and public cloud pricing.

5.1 Sensitivity to Researcher Delay Tolerance

Figure 6 show the impact of relaxing SLA constraints on systemic performance. By increasing the maximum allowable AI workload delay from 0 hours (baseline) to just 12 hours, the framework yields a drastic reduction in both required cloud outsourcing (MWh) and total operational cost. This confirms that even a minimal buffer enables the virtual queue to act as a computational shock absorber.

The framework reveals a counterintuitive, marginal rebound in operational costs as delay tolerance expands from 24 to 48 hours. Figure 7 isolates the cause of this variance by mapping the model's multi-objective behavior. In the highlighted "Trade-off Zone," the MPC algorithm actively sacrifices a slight increase in financial cost to secure a steeper, continued drop in grid carbon emissions. Because the objective function formally weights both cost and carbon, the solver's emphasis on carbon emissions outweighs the slight premium in off-peak electricity pricing.

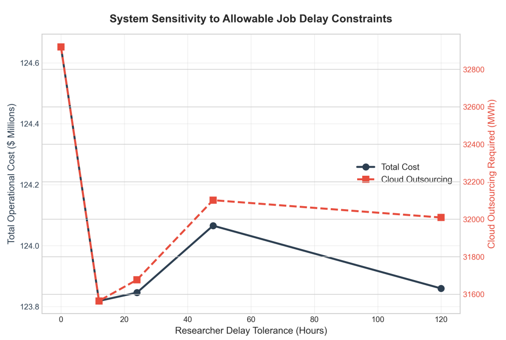

Figure 6: System Sensitivity to Allowable Job Delay Constraints.

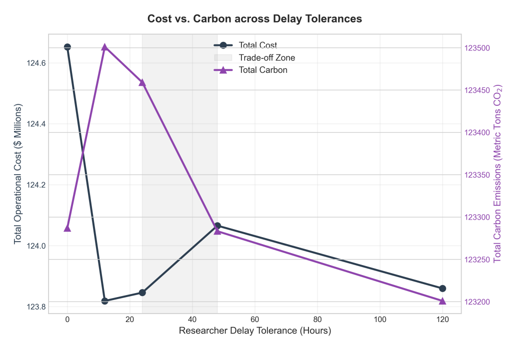

Figure 7: The Multi-Objective Trade-Off (Cost vs. Carbon across Delay Tolerances).

5.2 Sensitivity to Public Cloud Pricing

Because the high-performance computing market is subject to extreme pricing volatility, Figure 8 tests the framework's sensitivity to AWS spot pricing under a fixed 48-hour delay tolerance. When cloud capacity is relatively cheap ($600/MWh), the model freely outsources excess jobs to minimize local grid strain. However, under a severe scarcity scenario ($2,400/MWh), the solver determines that outsourcing is prohibitively expensive and forces jobs to run locally on the Stanford grid. This local computational bottleneck drives a sharp increase in total carbon emissions.

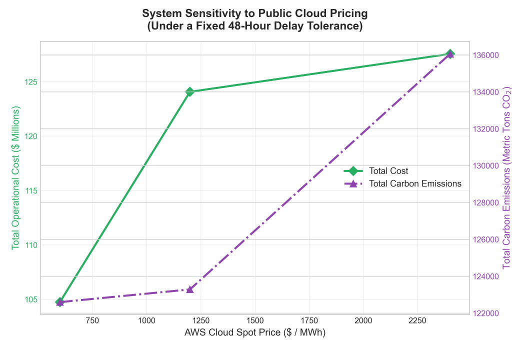

Figure 8: System Sensitivity to Public Cloud Pricing

6 Discussion

The results of this study challenge the pervasive industry idea that AI compute must operate as an inflexible load. As demonstrated by the MPC model's $7.21 million cost reduction, failing to leverage the inherent temporal flexibility of asynchronous AI training jobs constitutes a massive financial oversight. By introducing a five-day delay tolerance constraint, the university can entirely circumvent emergency cloud bursting fees.

Furthermore, the 2,926 MT reduction in carbon emissions highlights a critical avenue for sustainable computing. Currently, tech giants are pursuing decarbonization by constructing co-located renewables near data centers [12]. However, our digital twin proves that intelligent software scheduling can achieve real grid-level emission reductions without requiring new infrastructure construction.

We encountered several modeling challenges during our construction of the optimization problem. The most significant was the absence of Stanford-native GPU workload traces, which required us to substitute compute load with the Alibaba cluster trace and rescale it to match Stanford's 6 MW capacity constraint. Additionally, the omission of campus-wide inflexible loads (lighting, plug loads, hospital demand) prevented the model from capturing Stanford's full demand charge structure, which represents a meaningful component of real operational costs. Infeasibility events on isolated simulation days, caused by thermodynamic constraint conflicts at tank boundary conditions, were handled by carrying forward the prior state rather than re-optimizing, which introduces minor inaccuracies in the annual totals.

Future iterations of this model would address these limitations in three ways. First, integrating Stanford SRCF workload telemetry would replace the Alibaba proxy with campus-verified job traces. Second, incorporating total campus load profiles and a demand charge term in the cost objective would allow the optimizer to use thermal storage for peak shaving. Third, extending the MPC prediction horizon would improve the solver's ability to anticipate multi-day weather and pricing patterns.

Ultimately, this simulation verifies that flexible AI load with district thermal energy systems (e.g. Stanford CEF) is an economically and environmentally beneficial option for co-located AI compute demand.

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